



LINEAR ALGEBRA AND ITS APPLICATIONS

Swetha D S

Department of Science and Humanities

Unit-4

Orthogonalization , Eigen Values and Eigen Vectors

CLASS-3

THE GRAM-SCHMIDT ORTHOGONALIZATION

The Gram-Schmidt process:

- It is a process of converting linearly independent vectors into orthonormal vectors.
- Consider any 3 independent vectors a, b, c . Then the first orthonormal $q_1 = a/\text{norm}(a)$.
- If ' b ' is perpendicular to the vector ' a ' then $q_2 = b/\text{norm}(b)$ otherwise $B = b - (q_1^T b)q_1$ and $q_2 = B/\text{norm}(B)$.

- If 'c' is perpendicular to the plane spanned by the vectors a and b then $q_3 = c / \text{norm}(c)$
otherwise $C = c - (q_1^T c)q_1 - (q_2^T c)q_2$ and $q_3 = C / \text{norm}(C)$.

This is the one idea of the whole Gram-Schmidt process, to subtract from every new vector its components in the directions that are already settled. That idea is used over and over again. When there is a fourth vector, we subtract away its components in the directions of q_1, q_2, q_3 .

LINEAR ALGEBRA AND ITS APPLICATIONS

Problems:-

1. From the vectors a, b, c find orthonormal vectors q_1, q_2

& q_3 given $a = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, c = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

Ans: From Gram-Schmidt process $q_1 = \frac{a}{\|a\|} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$

$$\begin{aligned} q_2 &= \frac{e_2}{\|e_2\|} \quad \text{where } e_2 = b - [q_1^T b] q_1 \\ &= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} \\ &= \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix} \end{aligned}$$

$$\therefore q_2 = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right)$$

$$q_3 = \frac{e_3}{\|e_3\|}, \quad e_3 = (-[q_1^T c]q_1 - [q_2^T c]q_2)$$

$$e_3 = \begin{pmatrix} -2/3 \\ 2/3 \\ 2/3 \end{pmatrix}, \quad \|e_3\| = \frac{2\sqrt{3}}{3}$$

$$\therefore q_3 = \begin{pmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$



THANK YOU

SWETHA D S

Department of Science and Humanities